Homework 11

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Due 5/24/2025

Homework Instructions

Make sure to add your name to the header of the document. When submitting the assignment on Gradescope, be sure to assign the appropriate pages of your submission to each Exercise.

Note: This assignment has two parts. The R assignment should be submitted to the Homework 11, Part 1 assignment on Gradescope like normal. Answers with interpretations, numeric inputs, and multiple choice questions should be submitted to the Homework 11, Part 2 assignment in Gradescope. Interpretation questions should be submitted to the second Gradescope assignment rather than written in this assignment.

The points for each Exercise are noted in the title.

For questions that require code, please create or use the code chunk directly below the question and type your code there. Your knitted pdf will then show both the code and the output, so that we can assess your understanding and award any partial credit.

You are encouraged to knit your file as you work, to check that your coding and formatting are done so appropriately. This will also help you identify and locate any errors more easily.

Homework Setup

We'll use the following packages for this homework assignment. **Note: we haven't used the boot package before. To use this package, you'll want to run `install.packages(boot)` in the Console before knitting this document.** We'll also read in the `bechdel` & `bechdel_test` data from a csv file. To access the data, you'll want to download the dataset from Canvas, and place it in the same folder as this R Markdown document. You'll then be able to write code to load in the data.

```r
library(ggplot2)
library(boot)
library(faraway)
```

```
## Warning: package 'faraway' was built under R version 4.4.3
```

```
##
## Attaching package: 'faraway'
```

```
## The following objects are masked from 'package:boot':
##
##     logit, melanoma
```

Exercise 1: Formatting [5 points]

The first five points of the assignment will be earned for properly formatting your final document. Check that you have:

- included your name on the document
- properly assigned pages to exercises on Gradescope
- selected **page 1 (with your name)** and this page for this exercise (Exercise 1)
- all code is printed and readable for each question
- all output is printed
- generated a pdf file

Exercise 2: Air Quality Models [7 points]

For the first exercise on this assignment, we'll return to the built-in airquality dataset. Before starting our analyses, run the following line of code, which removes any observations with `NA`s and retains only complete observations.

```
airquality = na.omit(airquality)
```

The models fit in this question will be used on Question 1 of Gradescope.

part a

Fit a quadratic model to the data, predicting Ozone from the terms **Temp** and **Temp**^{2}. Use the `poly` method to create your terms. Print the resulting coefficients table.

```
airquality = na.omit(airquality)
model = lm(Ozone ~ poly(Temp, 2, raw = TRUE), data = airquality)
summary(model)$coefficients
```

```
##                                Estimate   Std. Error   t value     Pr(>|t|)
## (Intercept)                 292.95885406 123.92019134  2.364093 0.0198610793
## poly(Temp, 2, raw = TRUE)1   -9.22680345   3.25492644 -2.834719 0.0054759157
## poly(Temp, 2, raw = TRUE)2    0.07602414   0.02115988  3.592845 0.0004941148
```

part b

Now, fit a quartic model (with up to the fourth-order terms) that predicts Ozone from Temp. Again, use the `poly` function to create your terms.

Print the summary of the fitted coefficients.

```
quartic_model = lm(Ozone ~ poly(Temp, 4, raw = TRUE), data = airquality)
summary(quartic_model)$coefficients
```

```
##                                  Estimate   Std. Error   t value   Pr(>|t|)
## (Intercept)                 -1.468561e+04 6.389228e+03 -2.298495 0.02349568
## poly(Temp, 4, raw = TRUE)1   8.053131e+02 3.401294e+02  2.367667 0.01971401
## poly(Temp, 4, raw = TRUE)2 -1.637328e+01 6.735368e+00 -2.430941 0.01673756
## poly(Temp, 4, raw = TRUE)3  1.462188e-01 5.881207e-02  2.486204 0.01447259
## poly(Temp, 4, raw = TRUE)4 -4.828006e-04 1.910992e-04 -2.526440 0.01300022
```

part c

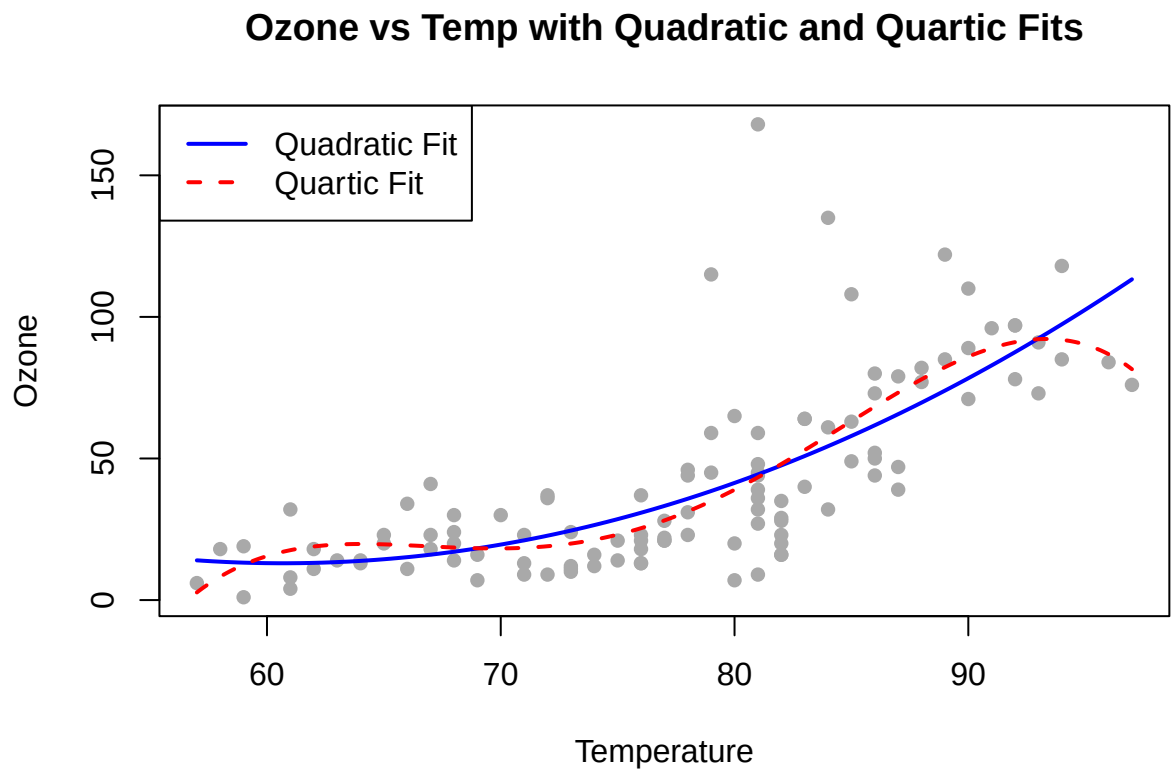
Create a scatterplot that visualizes the relationship between Ozone & Temp, where Ozone is our response variable. Add two summary lines to this plot: one that represents a quadratic (2nd order) relationship and one that represents a quartic (4th order) relationships. (Note: you may make two different plots for this question, one for each model, if you'd prefer.)

```r
plot(airquality$Temp, airquality$Ozone,
     main = "Ozone vs Temp with Quadratic and Quartic Fits",
     xlab = "Temperature",
     ylab = "Ozone",
     pch = 16, col = "darkgray")

quad_model = lm(Ozone ~ poly(Temp, 2, raw = TRUE), data = airquality)
temp_seq = seq(min(airquality$Temp), max(airquality$Temp), length.out = 300)
quad_pred = predict(quad_model, newdata = data.frame(Temp = temp_seq))
lines(temp_seq, quad_pred, col = "blue", lwd = 2)

quartic_model = lm(Ozone ~ poly(Temp, 4, raw = TRUE), data = airquality)
quartic_pred = predict(quartic_model, newdata = data.frame(Temp = temp_seq))
lines(temp_seq, quartic_pred, col = "red", lwd = 2, lty = 2)

legend("topleft",
       legend = c("Quadratic Fit", "Quartic Fit"),
       col = c("blue", "red"),
       lty = c(1, 2),
       lwd = 2)
```



Exercise 3: Understanding Logistic Regression [0 points]

All components of questions 2 & 3 on Gradescope can be completed on Gradescope. You do not need to include any work on this assignment, unless you choose to use R as a calculator below.

```
# Use this code chunk for your calculation, if needed.
```

The exercise is used as a placeholder, so that the question number of Gradescope should match the corresponding Exercise in Part 1 for the remainder of the assignment.

Exercise 4: Bechdel Test, Understanding a Model [2 points]

For this question, we will use data generated from the Bechdel test being applied to movies. The Bechdel test, named for a cartoonist, is a measure of female representation in fiction. To pass the test, the work must pass three rules:

- It must have two named female characters.

- The female characters have to talk to each other.
- The female characters must talk about something other than a male character.

We will use the `bechdel` dataset for Exercises 4-7 consisting of 216 movies released between 1971-2013. Variables in this dataset include the movie's Bechdel test rating along with other characteristics.

The Bechdel cartoon strip was first printed in 1985. Is there a relationship between the publication and later popularity of the strip and whether a movie passes the Bechdel test?

Fit a logistic regression model below, predicting whether the movie `pass`es the Bechdel test from the release `year` of the move. Print the summary of the model.

```r
bechdel = read.csv("bechdel.csv")
model = glm(pass ~ year, data = bechdel, family = binomial)
summary(model)
```

```
##
## Call:
## glm(formula = pass ~ year, family = binomial, data = bechdel)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -85.54997   36.28159  -2.358   0.0184 *
## year          0.04263    0.01811   2.354   0.0186 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 298.25  on 215  degrees of freedom
## Residual deviance: 292.22  on 214  degrees of freedom
## AIC: 296.22
##
## Number of Fisher Scoring iterations: 4
```

Exercise 5: Bechdel Test, A Second Model [6 points]

part a

Fit an additive logistic regression model to the Bechdel data, using `pass` as the response variable and `year`, `budget_2013`, `domgross_2013`, `intgross_2013`, `rated`, `metascore`, `imdb_rating`, and `runtime` as predictor variables. Print the summary of this model.

```r
model2 = glm(pass ~ year + budget_2013 + domgross_2013 + intgross_2013 +
             rated + metascore + imdb_rating + runtime,
             data = bechdel, family = binomial)

summary(model2)
```

```
##
## Call:
## glm(formula = pass ~ year + budget_2013 + domgross_2013 + intgross_2013 +
##     rated + metascore + imdb_rating + runtime, family = binomial,
##     data = bechdel)
##
## Coefficients:
##                 Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -4.195e+01  4.511e+01  -0.930 0.352425
## year           2.214e-02  2.237e-02   0.990 0.322298
## budget_2013   -1.472e-08  4.120e-09  -3.573 0.000353 ***
## domgross_2013 -4.390e-09  3.878e-09  -1.132 0.257554
## intgross_2013  2.215e-09  1.768e-09   1.253 0.210183
## ratedPG       -4.057e-01  1.192e+00  -0.340 0.733570
## ratedPG-13    -1.648e-01  1.166e+00  -0.141 0.887636
## ratedR        -6.487e-01  1.182e+00  -0.549 0.583261
## metascore      1.048e-02  1.258e-02   0.833 0.404918
## imdb_rating   -5.853e-01  2.490e-01  -2.351 0.018735 *
## runtime        1.838e-02  9.851e-03   1.866 0.062049 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 298.25  on 215  degrees of freedom
## Residual deviance: 269.36  on 205  degrees of freedom
## AIC: 291.36
##
## Number of Fisher Scoring iterations: 4
```

part b

Perform a likelihood ratio test, comparing the logistic model fit in Exercise 5a to the logistic model fit in Exercise 4.

```r
model1 = glm(pass ~ year, family = binomial, data = bechdel)
model2 = glm(pass ~ year + budget_2013 + domgross_2013 + intgross_2013 +
             rated + metascore + imdb_rating + runtime,
             family = binomial, data = bechdel)

anova(model1, model2, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: pass ~ year
## Model 2: pass ~ year + budget_2013 + domgross_2013 + intgross_2013 + rated +
##     metascore + imdb_rating + runtime
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1       214     292.22
## 2       205     269.36  9   22.864 0.006508 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Answer: The second model seems to be a better fit, since the p-value is below the alpha level of 0.01 as indicated on Gradescope, and it shows that at least one of the additional variables that was added helps improve the model. Plus, the first model did not have sufficient evidence to reject the null hypothesis.

Exercise 6: Bechdel Test, Model Selection [6 points]

part a

Perform model selection using a stepwise approach, the additive model from Exercise 5a as the largest considered model, and BIC as the metric.

```
full_model = glm(pass ~ year + budget_2013 + domgross_2013 + intgross_2013 +
                 rated + metascore + imdb_rating + runtime,
                 family = binomial, data = bechdel)

n = nrow(bechdel)
stepwise_model = step(full_model, direction = "both", k = log(n))
```

```
## Start:  AIC=328.49
## pass ~ year + budget_2013 + domgross_2013 + intgross_2013 + rated +
##     metascore + imdb_rating + runtime
##
##                 Df Deviance    AIC
## - rated          3   271.35 314.35
## - metascore      1   270.06 323.81
## - year           1   270.35 324.10
## - domgross_2013  1   270.69 324.44
## - intgross_2013  1   270.96 324.71
## - runtime        1   272.87 326.62
## <none>               269.36 328.49
## - imdb_rating    1   275.21 328.97
## - budget_2013    1   284.33 338.09
##
## Step:  AIC=314.35
## pass ~ year + budget_2013 + domgross_2013 + intgross_2013 + metascore +
##     imdb_rating + runtime
##
##                 Df Deviance    AIC
## - metascore      1   272.09 309.71
## - domgross_2013  1   272.40 310.03
## - intgross_2013  1   272.82 310.44
## - year           1   273.02 310.65
## - runtime        1   274.85 312.48
## <none>               271.35 314.35
## - imdb_rating    1   278.28 315.91
## - budget_2013    1   284.77 322.39
## + rated          3   269.36 328.49
##
## Step:  AIC=309.71
```

```
## pass ~ year + budget_2013 + domgross_2013 + intgross_2013 + imdb_rating +
##     runtime
##
##                 Df Deviance    AIC
## - domgross_2013  1   272.97 305.22
## - intgross_2013  1   273.36 305.61
## - year           1   273.88 306.13
## - runtime        1   275.62 307.88
## <none>               272.09 309.71
## - imdb_rating    1   279.44 311.69
## + metascore      1   271.35 314.35
## - budget_2013    1   285.59 317.84
## + rated          3   270.06 323.81
##
## Step:  AIC=305.22
## pass ~ year + budget_2013 + intgross_2013 + imdb_rating + runtime
##
##                 Df Deviance    AIC
## - intgross_2013  1   273.42 300.30
## - year           1   276.23 303.11
## - runtime        1   276.41 303.29
## <none>               272.97 305.22
## - imdb_rating    1   280.26 307.14
## + domgross_2013  1   272.09 309.71
## + metascore      1   272.40 310.03
## - budget_2013    1   285.78 312.66
## + rated          3   271.19 319.57
##
## Step:  AIC=300.3
## pass ~ year + budget_2013 + imdb_rating + runtime
##
##                 Df Deviance    AIC
## - year           1   276.26 297.76
## - runtime        1   277.02 298.52
## <none>               273.42 300.30
## - imdb_rating    1   280.27 301.77
## + metascore      1   272.86 305.12
## + intgross_2013  1   272.97 305.22
## + domgross_2013  1   273.36 305.61
## - budget_2013    1   288.46 309.96
## + rated          3   271.46 314.46
##
## Step:  AIC=297.76
## pass ~ budget_2013 + imdb_rating + runtime
##
##                 Df Deviance    AIC
## - runtime        1   279.69 295.82
## <none>               276.26 297.76
## + year           1   273.42 300.30
## - imdb_rating    1   286.02 302.14
## + metascore      1   275.71 302.58
## + domgross_2013  1   276.06 302.94
## + intgross_2013  1   276.23 303.11
## - budget_2013    1   291.85 307.98
```

```
## + rated          3   273.63 311.26
##
## Step:  AIC=295.82
## pass ~ budget_2013 + imdb_rating
##
##                 Df Deviance    AIC
## <none>               279.69 295.82
## - imdb_rating    1   286.06 296.81
## + runtime        1   276.26 297.76
## + year           1   277.02 298.52
## + metascore      1   279.11 300.61
## + domgross_2013  1   279.61 301.11
## + intgross_2013  1   279.62 301.12
## - budget_2013    1   291.98 302.73
## + rated          3   277.08 309.33
```

```r
summary(stepwise_model)
```

```
##
## Call:
## glm(formula = pass ~ budget_2013 + imdb_rating, family = binomial,
##     data = bechdel)
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)  2.883e+00  1.037e+00   2.781  0.00542 **
## budget_2013 -9.515e-09  2.913e-09  -3.267  0.00109 **
## imdb_rating -3.608e-01  1.472e-01  -2.452  0.01422 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 298.25  on 215  degrees of freedom
## Residual deviance: 279.70  on 213  degrees of freedom
## AIC: 285.7
##
## Number of Fisher Scoring iterations: 4
```

Answer:

part b

Perform a likelihood ratio test, comparing the additive model from Exercise 5a to the stepwise model selected in Exercise 6a.

```r
model_additive = glm(pass ~ year + budget_2013 + domgross_2013 + intgross_2013 +
                     rated + metascore + imdb_rating + runtime,
                     family = binomial, data = bechdel)
model_stepwise = glm(pass ~ budget_2013 + imdb_rating,
                     family = binomial, data = bechdel)
anova(model_stepwise, model_additive, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: pass ~ budget_2013 + imdb_rating
## Model 2: pass ~ year + budget_2013 + domgross_2013 + intgross_2013 + rated +
##     metascore + imdb_rating + runtime
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1       213     279.69
## 2       205     269.36  8   10.338   0.2421
```

Answer:

Exercise 7: Bechdel Test, Model Interpretations [0 points]

For this Exercise, we will analyze a logistic regression model with `pass` as the response variable and `imdb_rating` and `budget_2013` as the predictors.

You may use this portion of your assignment to perform any calculations that you need, although nothing needs to be shown here.

```r
model = glm(pass ~ budget_2013 + imdb_rating, family = binomial, data = bechdel)
new_movie = data.frame(
  budget_2013 = 60000000,
  imdb_rating = 6.7
)
pred = predict(model, newdata = new_movie, type = "link", se.fit = TRUE)
logit_fit = pred$fit
logit_se = pred$se.fit

z = qnorm(0.975)  # 1.96 for 95% CI
lower_logit = logit_fit - z * logit_se
upper_logit = logit_fit + z * logit_se

invlogit = function(x) {
  exp(x) / (1 + exp(x))
}

lower_prob = invlogit(lower_logit)
upper_prob = invlogit(upper_logit)
fit_prob   = invlogit(logit_fit)
cat("Estimated probability:", round(fit_prob, 4), "\n")
```

```
## Estimated probability: 0.4738
```

```r
cat("95% CI: [", round(lower_prob, 4), ",", round(upper_prob, 4), "]\n")
```

```
## 95% CI: [ 0.4044 , 0.5442 ]
```

Exercise 8: Evaluating a Bechdel Classifier [3 points]

For this question, we will use the model from Exercise 7 as the basis for a classifier that predicts whether a movie passes or doesn't pass the Bechdel test.

For this exercise, we will examine the performance of our classifier on the `bechdel_test` data from Canvas. Note that the previous models have not previously seen the `bechdel_test` data before.

part a

Calculate the predicted probabilities of passing the Bechdel test for our test data based on the model from Exercise 7. Save these predicted probabilities.

```
bechdel_test = read.csv("bechdel_test.csv")
head(bechdel_test)
```

```
##   year              title budget_2013 intgross_2013 rated imdb_rating runtime
## 1 2003         Wrong Turn    12660870      36272829     R         6.1      84
## 2 2009              Gamer    54286580      38068663     R         5.8      95
## 3 2002 Rabbit-Proof Fence     9066255      21845695    PG         7.5      94
## 4 2011                Rio    93212171     504919774     G         7.0      96
## 5 2011             Priest    62141447      87157549 PG-13         5.7      87
## 6 2005       Coach Carter    53686491      91469619 PG-13         7.2     136
##   pass
## 1    1
## 2    0
## 3    1
## 4    0
## 5    0
## 6    1
```

```
bechdel_test$predicted_prob = predict(model, newdata = bechdel_test, type = "response")
head(bechdel_test$predicted_prob)
```

```
## [1] 0.6368990 0.5680980 0.5227378 0.3706926 0.5585884 0.4439086
```

part b

Generate a simple classifier that predicts a movie will pass the Bechdel test if it's predicted probability is greater than (or equal to) 0.7 and that a movie will not pass the Bechdel test if it's predicted probability is less than 0.7.

Record the predicted classes for the simple classifier.

```
bechdel_test$predicted_class = ifelse(bechdel_test$predicted_prob >= 0.7, 1, 0)
head(bechdel_test[, c("predicted_prob", "predicted_class")])
```

```
##   predicted_prob predicted_class
## 1      0.6368990               0
## 2      0.5680980               0
## 3      0.5227378               0
```

```
## 4      0.3706926               0
## 5      0.5585884               0
## 6      0.4439086               0
```

```r
table(bechdel_test$pass, bechdel_test$predicted_class)
```

```
##
##       0   1
##   0 277   6
##   1 211  13
```

```r
table(bechdel_test$pass)
```

```
##
##   0   1
## 283 224
```

```r
rate_to_beat = max(prop.table(table(bechdel_test$pass)))
rate_to_beat
```

```
## [1] 0.5581854
```